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$$\begin{aligned}\int_1^2 \frac{1}{x\sqrt{x^2-1}} dx & \stackrel{x=\frac{1}{\cos t}}{=} \int_0^{\frac{\pi}{3}} \frac{1}{\frac{1}{\cos t} \cdot \frac{\sin t}{\cos t}} \cdot \frac{\sin t}{\cos t} dt = \int_0^{\frac{\pi}{3}} \frac{\sin t}{\sin t} dt \\ & = [t]_0^{\frac{\pi}{3}} = \frac{\pi}{3}\end{aligned}$$

References